



EV fast-charging cybersecurity risks exposed

Southwest Research Institute (SwRI) engineers have found cybersecurity vulnerabilities in electric vehicles (EVs) using direct current fast-charging systems, the fastest common charging method. This high-voltage technology uses power



SwRI developed an adversary-in-the-middle device with a modified combined charging system to test the cyber resiliency of ISO 15118-compliant vehicle-to-grid, direct current fast charging systems. The high-voltage technology relies on power line communications (PLC) to transmit smart-grid data between vehicles and charging equipment (Credit: Southwest Research Institute)

line communication (PLC) to transmit smart-grid data. SwRI exploited PLC vulnerabilities in the lab to access network keys and digital addresses on chargers and vehicles. The team developed an adversary-in-the-middle (AitM) device with specialised software, allowing eavesdropping on EV and charging

station communications. They identified network membership keys, enabling devices to join the network and monitor traffic. SwRI recommends adding encryption to these keys for security, though it presents challenges. Future research will focus on zero-trust architectures for EV cybersecurity.

Magnetic tunnel device cuts AI energy use

Researchers at the University of Minnesota Twin Cities have developed an advanced hardware device that could reduce energy consumption in AI computing by over 1000 times. As AI demand grows, enhancing energy efficiency while maintaining performance and low costs is crucial. AI processes consume substantial power due to data transfer between logic and memory. The new approach, computational random-access memory (CRAM), keeps data within memory, using magnetic tunnel junctions (MTJs) for efficient computation. This method eliminates the energy-intensive data transfers of traditional architectures, using spintronic devices that store data via electron spin, offering superior energy efficiency, speed, and resilience.

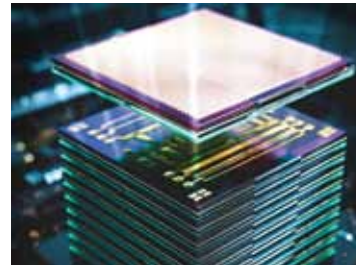
E-skins detect liquid movement on robots

Recent research aims to develop electronic skins for robots, smart prosthetics, and bio-inspired systems to detect textures and tactile qualities. A team from Donghua University

and the National University of Singapore created an electronic skin (e-skin) that detects surface liquid movement by converting sliding droplet motion into electrical signals. Inspired by a soccer ball rolling on tiles, the dynamic electronic skin uses interwoven electrodes to sense and transmit droplet behaviour. This technology enhances humanoid robots' tactile sensing, potentially improving performance in tasks involving moisture and narrowing the sensory gap between humans and robots.

First 10-stack 3D transistors for compact devices

Researchers at King Abdullah University of Science and Technology (KAUST) have developed three-dimensional (3D) integrated transistors that can be vertically stacked, achieving



Artistic illustration of the 10-stacked metal oxide transistors (Credit: Yuvaraja et al)

the highest number of stacked transistors to date. This design boosts performance and efficiency, addressing the limitations of conventional planar transistors. The team's 10-stack nanometre-thick compound semiconductor transistors feature

low thermal budgets and low interfacial roughness, utilising indium oxide (In₂O₃) and parylene-C for insulation. These transistors demonstrated remarkable performance, with applications in mobile devices, IoT, AI, machine learning, consumer electronics, and automotive electronics. Future research aims to scale down to the nanoscale, further improving power consumption, reliability, and stability.

Solid-state batteries boost longevity and density

Along with international collaborators, researchers at the Qingdao Institute of Bioenergy and Bioprocess Technology (QIBEBT) of the Chinese Academy of Sciences have introduced a cathode homogenisation strategy for all-solid-state lithium batteries (ASLBs). This approach enhances ASLBs' life cycle and energy density by using the zero-strain material Li_{1.75}Ti₂(Ge_{0.25}P_{0.75}S_{3.8}Se_{0.2})₃ (LTG_{0.25}PSSe_{0.2}). This material ensures efficient charge transport without additional additives, achieving over 20,000 stable cycles and a high energy density of 390 Wh kg⁻¹. By eliminating inactive additives, this strategy improves energy density and extends battery life, marking a significant advancement in energy storage technology with potential applications in electric vehicles and large-scale storage systems.